



MRINAL BIJU

B.Tech. - Computer Science & Engineering

Ph: +91-7306817636

Email: mrinalbiju.b23cs1146@mbcet.ac.in

Thiruvananthapuram, Kerala, India - 695316

LinkedIn: <https://www.linkedin.com/in/mrinal-biju-17126b274/>



PROFESSIONAL SUMMARY

Final-year B.Tech Computer Science student at Mar Baselios College of Engineering and Technology, Trivandrum, with a strong focus on AI/ML and full-stack application development. I enjoy building practical, end-to-end products — from SafeShelf, a React + Capacitor allergen-detection app, to a Computer Vision-based Crowd Panic & Stampede Risk Detection system using CSRNet, optical flow, and gradient boosting classification.

I work across Python, Java, Flutter, React, and Node.js, and I lean heavily on AI-assisted development tools like Claude Code and Google AI Studio to build and ship faster. Driven by curiosity and a problem-solving mindset, I'm looking to contribute to teams building impactful, intelligent software.

KEY EXPERTISE

Python Java C Flutter ReactJS React Native NodeJS GIT GitHub FastAPI OpenCV LangChain

EDUCATION

Mar Baselios College of Engineering and Technology (Autonomous)

2023 - 2027

B.Tech. - Computer Science & Engineering | CGPA: 7.26 / 10

St. Thomas Central School, Thiruvananthapuram

2023

12th | CBSE | Percentage: 72 / 100

St. Thomas Central School, Thiruvananthapuram

2021

10th | CBSE | Percentage: 80 / 100

INTERNSHIPS

UST | Retailing

20 May, 2026 - 30 Jun, 2026

Software Engineer Intern

SafeShelf — AI-powered mobile grocery companion that scans products and analyzes ingredients against a user's dietary constraints.

Built a cross-platform mobile app (iOS/Android) with React 19, TypeScript, Capacitor, and TanStack Router/Query, featuring real-time barcode and camera-based product scanning.

Developed a FastAPI (Python) backend with SQLAlchemy and async PostgreSQL, exposing versioned REST APIs for product search, cart, dietary profiles, and suitability analysis.

Integrated an LLM (Groq) and the USDA FoodData Central API to analyze product ingredients and flag dietary conflicts in real time.

Implemented secure JWT authentication with access/refresh token rotation and httpOnly cookies.

Set up the full pnpm/uv monorepo dev workflow — concurrent dev servers, mkcert HTTPS for on-device camera testing, Biome/ruff linting, and Docker deployment.

PROJECTS

SafeShelf

20 May, 2026 - 30 Jun, 2026

Mentor: Anand Harisankar | **Team Size:** 5

Key Skills: React TypeScript Python FastAPI PostgreSQL REST Tailwind CSS GIT

SafeShelf is an AI-powered mobile grocery companion that scans grocery products and analyzes their ingredients against a user's personal dietary constraints (allergies, restrictions, and nutrition goals), flagging unsuitable items in real time.

I worked on building the application end-to-end across the mobile frontend and backend. On the frontend, I developed a cross-platform (iOS/Android) app using React 19, TypeScript, and Capacitor, implementing real-time barcode and camera-based product scanning along with cart management and dietary-profile screens. On the backend, I built a FastAPI service with SQLAlchemy and async PostgreSQL, exposing versioned REST APIs for product search, suitability analysis, user profiles, and authentication.

A core part of my work was integrating a large language model (Groq) with the USDA FoodData Central nutrition database to analyze product ingredients and automatically detect conflicts with each user's dietary requirements. I also implemented secure JWT-based authentication with access/refresh token rotation, and set up the project's development workflow including a pnpm/uv monorepo, HTTPS for on-device camera testing, linting, and Docker-based deployment.

Outcome: Delivered a functional full-stack application that lets users make safer, faster grocery decisions by instantly identifying whether a product fits their dietary needs.

Crowd Panic & Stampede Risk Detection System

01 Jan, 2026 - 30 Apr, 2026

Mentor: Ms Poorna BR | **Team Size:** 4

Key Skills: OpenCV Python

Built a crowd density and motion analysis pipeline using CSNet and Farneback Optical Flow to detect panic patterns from video, validated on

ShanghaiTech and UMN datasets. Engineered a temporal feature pipeline with sliding windows; trained a Gradient Boosting model for real-time stampede risk classification and early prediction. Deployed a Flask dashboard with live annotated feed, heatmaps, and an early warning alert system for security personnel

Musify

30 Dec, 2024 - 11 Feb, 2025

Team Size: 3

Key Skills: Dart C++

Developed a cross-platform music streaming application frontend inspired by Spotify using Flutter and Dart. Designed responsive user interfaces, implemented music browsing and playback features, navigation, playlists, and a modern user experience while gaining hands-on experience in mobile app development and UI/UX design.

SEMINARS / TRAININGS / WORKSHOPS

Flutter Workshop Institute Name: ICT Academy

30 Dec, 2024 - 16 Feb, 2025

Key Skills: Dart App Development

Participated in a hands-on Flutter workshop focused on cross-platform mobile application development. Learned the fundamentals of Flutter, Dart programming, widget-based UI design, state management concepts, navigation, and app deployment. Developed sample mobile applications to gain practical experience in building responsive and efficient Android and iOS applications from a single codebase.

Customized AI Kitchen

01 May, 2024 - 30 Jun, 2024

Institute Name: Intel

Key Skills: Python

Successfully completed Intel Unnati Industrial Training 2024, working on the project "Customized AI Kitchen for India" under the guidance of industry and academic mentors. Gained hands-on experience in Artificial Intelligence, problem-solving, data-driven application development, and collaborative project execution while developing AI-based solutions tailored to Indian culinary requirements.

WEB LINKS

- Github - <https://github.com/mrinalbiju>

PERSONAL DETAILS

Gender: Male

Marital Status: Single

Current Address: Flat no. 6A, SFS Orchard, Pallipuram,
Thiruvananthapuram, Kerala, India - 695316

Emails: mrinalbiju.b23cs1146@mbcet.ac.in , mrinalbiju81@gmail.com

Date of Birth: 22 Aug, 2005

Known Languages: Malayalam(Fluent, Native), English(Fluent),
Hindi

Permanent Address: Flat no. 633, EMS Nagar, Pattoor,
Vanchiyoor, Thiruvananthapuram, Kerala, India - 695035

Phone Numbers: +91-7306817636, +91-9496154551